

# Maxillary & mandibular canines

|                                 | Maxillary canine                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Mandibular canine                                                                                                                                                                                                                                                                                                        |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enamel organ appearance         | 6 MIUL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 6 MIUL                                                                                                                                                                                                                                                                                                                   |
| Beginning of calcification      | 4-5 months old                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4-5 months old                                                                                                                                                                                                                                                                                                           |
| Crown completion                | 6-7 years old                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 6-7 years old                                                                                                                                                                                                                                                                                                            |
| Eruption                        | 11-12 years old                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 9-10 years old                                                                                                                                                                                                                                                                                                           |
| Root completion                 | 14-15 years old                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 12-14 years old                                                                                                                                                                                                                                                                                                          |
| Mesial outline                  | <b>convex</b> till the incisal/middle junction (the contact area). Then become <b>concave</b> till cusp tip                                                                                                                                                                                                                                                                                                                                                                           | <b>Straight</b> till the mesial contact area which's close to the mesioincisal angle.                                                                                                                                                                                                                                    |
| Distal outline                  | <b>concave</b> till the <b>center of middle third</b> (the contact area) then <b>convex</b> to the cusp tip                                                                                                                                                                                                                                                                                                                                                                           | <b>Convex</b> till the <b>incisal/middle</b> junction<br>*it's shorter than the mesial outline as the cusp ridge slopes downwards.                                                                                                                                                                                       |
| Labial outline (proximal view)  | <b>Convex</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Less <b>convex</b>                                                                                                                                                                                                                                                                                                       |
| Lingual outline (proximal view) | Cervical third: <b>convex</b><br>Middle third: <b>straight</b><br>Incisal third: <b>concave</b> (cusp tip)                                                                                                                                                                                                                                                                                                                                                                            | Similar but <b>smaller curvature</b> with a <b>concavity</b> in the middle third (due to the absence of complete lingual ridge)                                                                                                                                                                                          |
| The root                        | <b>•labial &amp; lingual views:</b><br>- It is long, slender, conical in shape with distal curvature of the apical third & a pointed apex.<br><b>•proximal view:</b><br>-exhibits a developmental depression at most of its length (deeper distally)<br>-may curve <b>labially</b>                                                                                                                                                                                                    | <b>•labial &amp; lingual views:</b><br>-Same with more pointed apex<br><br><b>•proximal view:</b><br>-More marked developmental depression<br>-May be <b>bifurcated</b> at the apical third                                                                                                                              |
| Pulp cavity (chamber & canal)   | <b>•mesiodistal section:</b><br>- Has narrow pulp chamber pointed incisally<br>-The root canal is narrow<br><b>•labiolingual section:</b><br>-Narrow pulp chamber pointed incisally.<br>-The root canal starts wide in cervical & middle 1/3s then decreases in size in the apical 1/3 until becomes constricted at apical foramen.<br>-Larger apical foramen than those of incisors<br><b>•cross section at the cervix:</b><br>-Oblong (rectangle or oval) canal<br>- lab-ling > M-D | <b>• Pulp chamber &amp; root canal are similar, but smaller</b><br><b>• The root may have 2 root canals</b> (1 labial & 1 Lingual) open separately in 2 apical foramina or get united at root apex & open in single apical foramen<br><b>•the root may be bifurcated &amp; in this case the 2 canals open separately</b> |
| features                        | -Labial ridge (over developed middle lobe)<br>-Mesiolabial & distolabial developmental depressions<br>-Mesial & distal lingual fossae sectioned by the lingual ridge<br>-the cusp tip is inclined <b>labially &amp; mesially</b><br>-Bulky crown with <b>greater lab-ling measurement than any of anterior teeth.</b><br>-MD width < Lab-Ling diameter                                                                                                                                | -less prominent elevations & depressions<br><br>-the lingual ridge exists in the incisal third only, exhibiting just one lingual fossa<br>- the cusp tip is <b>centered</b> (inclined <b>lingually</b> in some cases) & shifted <b>mesially</b><br><br>- MD width < Lab-Ling diameter                                    |